
1. One-paragraph summary

GEO-INTEL takes a natural-language environmental question (for example, "*Find areas around Dehradun where urban expansion is associated with vegetation loss between 2018 and 2025*") and converts it into a validated remote-sensing and GIS workflow. The LLM only **plans** and **writes**. Every number comes from deterministic code (Sentinel-2 processing, classification, change detection, GIS statistics). Literature evidence comes from retrieval (RAG). The system never claims causation; it reports spatial and temporal **association**.

Status at midterm: the architecture, definitions, data pipeline, classifiers, change-detection and spatial-statistics modules, validators, mock agent and demo app are **built and tested on synthetic data**. On real data we have **verified** access to Sentinel-2 imagery, the scene inventory for both epochs, and a first reflectance-harmonisation check. The real composites, labelled reference data, accuracy figures and change results are **NOT YET RUN**.

2. Scope and fixed design decisions

Item	Decision
Workflow	One only: urban expansion vs. vegetation loss
Study area	Approved 30 km × 30 km square around Dehradun (about 900 km ²), bounds 77.9127°E–78.2323°E, 30.1465°N–30.4238°N
Epochs	T1 = Nov 2018–Mar 2019, T2 = Nov 2024–Mar 2025 (dry season, to avoid monsoon clouds)
Imagery	Sentinel-2 L2A, 10 m, free public catalogue (Earth Search COGs, no credentials)
Compute CRS	EPSG:32644 (UTM 44N). Areas and distances are never computed in degrees
Land-cover classes	5: built-up, tree, cropland/grass, bare, water. Priority order: water > built-up > tree > cropland/grass > bare > unclassified
Classifiers	Rule-based baseline and Random Forest. SegFormer is postponed
Change method	Post-classification transition matrix plus an NDVI-difference baseline
"Associated"	Both change types occur in the same 1 km grid cell, tested with Spearman correlation, Moran's I and Getis-Ord Gi*

Not in scope GraphRAG, multi-agent frameworks, deep-learning change detection, flood/road use case, React frontend

Claim policy. The system must say "associated with", "co-occurs with" or "spatially overlaps with". It must not say "caused", "led to", "due to", "responsible for" or "will flood". A phrase-based linter enforces this. The linter can miss novel wording, so human review is still needed.

3. Architecture

User question

- > Planner (LLM, temperature 0, structured JSON plan)
- > Plan validator (allow-listed tools, place inside AOI, years available)
- > Executor (deterministic tools, logged trace)
 - data fetch -> cloud mask -> composites -> indices -> classification
 - > change detection -> 1 km grid statistics
- > RAG retrieval (chunk-ID citations)
- > Report writer (numbers only from result JSON)
- > Output validator (number provenance, causal-language linter, citation check)
- > Map + report (Streamlit)

Separation of duties: the LLM never computes areas, distances or statistics. A "number provenance" check confirms every number in the report exists in the tool-output JSON, with matching units.

4. What was done, phase by phase

Phase 1: Problem definition and setup (BUILT)

- Repository, configuration file, PostGIS schema, Makefile, .gitignore, .env.example.
- Documents: problem statement with research questions and metric plan, formal definitions, gap table versus related work (GeoChat, SegFormer as a component, Shafique et al. 2022, GeoAgentic-RAG, plus LLM-Geo, GeoGPT, Change-Agent and similar), and a candidate paper list.
- **Open:** all 20 candidate papers are **UNVERIFIED**. Each must be checked by a person on Google Scholar or OpenAlex before it is cited.

Phase 2: Data pipeline (BUILT; real composites NOT YET RUN)

- STAC search and scene retrieval, Scene Classification Layer (SCL) cloud and shadow masking, median compositing into 10 m Cloud-Optimized GeoTIFFs, OSM layers, Copernicus 30 m DEM with slope and aspect (aspect stored as sine and cosine).
- Reflectance normalisation: scale 0.0001, with the post-January-2022 baseline offset handled per scene.
- Chunked processing with retries and partial COG output, so one failed read does not lose the run.
- Hosted run scripts for Colab or Kaggle, because rasterio cannot be imported on the development Windows machine (blocked by an Application Control policy).

Phase 3: GIS library and PostGIS (BUILT)

- CRS-checked functions: reproject, clip, area, length, buffer, overlay, zonal statistics, 1 km fishnet, raster-to-grid aggregation.

- Five-class scheme restored (an earlier version had silently merged to four).
- Joint valid-pixel mask for the transition matrix, reporting excluded area in km² and %.
- PostGIS schema and three documented spatial queries. The check script has **NOT YET RUN**.

Phase 4: Labels and accuracy tooling (BUILT; no labels exist)

- Seeded stratified candidate sampling, QGIS-ready GeoPackage, label schema (two annotators, confidence, notes).
- Confusion matrix, overall accuracy, per-class precision/recall/F1, Cohen's kappa between annotators, spatial block holdout to avoid leakage.
- The Random Forest currently trains on rule-based pseudo-labels. That is **circular** and is treated as bootstrap only. Real accuracy needs manual reference labels.

Phase 5: Change detection and association (BUILT; real results NOT YET RUN)

- 3×3 majority filter and minimum mapping unit (0.5 ha), NDVI-difference baseline.
- 1 km grid association: Spearman correlation, global Moran's I, Getis-Ord Gi* hotspots, top-N ranking.
- External consistency checker against GHSL, Hansen Global Forest Change and ESA WorldCover. Tiles needed: Hansen 40N_070E ; WorldCover N30E075 and N30E078 .
- GHSL epochs (2020, 2025) do not match T1/T2, and 2025 is a projection. These are rough consistency context only, never ground truth.
- Optional area-adjusted estimates with confidence intervals, which require labels.

Phase 6: Agent, RAG and validators (BUILT; live LLM NOT MEASURED)

- Mock planner, plan allow-list, in-memory retriever with chunk IDs, number-provenance checker, causal-language linter.
- 40-query benchmark with gold plans stored separately from planner prompts, plus adversarial tests (rounding, unit mix-ups, percentages, years, phrases such as "due to").
- The fixture benchmark scored 1.00, but the mock planner is scored against its own cached plans. That checks the plumbing, **not agent accuracy**. The real LLM planner is **NOT YET MEASURED**.

Phase 7: Demo and documentation (BUILT; app launch NOT YET VERIFIED)

- Streamlit app with a prominent "DEMO DATA" banner, running on synthetic fixtures only.
- Documents: demo script, Mermaid architecture, limitations, status table.

5. Real-data work on Colab

All results in this section come from executed runs.

5.1 Environment (VERIFIED)

Python 3.13.15, GDAL 3.8.4, rasterio 1.5.1, numpy 2.1.3, xarray 2026.9.0, Linux. The pipeline was written for Python 3.11; the exact-version checks were relaxed to "3.11 or newer".

5.2 Scene inventory (VERIFIED)

T1

T2

Items returned	44	49
Unique tile-date acquisitions	36	46
Tiles	44RKU (UTM 44) and 43RGP (UTM 43)	same
Tile-level cloud cover	median 1.1%, max 26.7%	median 0.2%, max 27.8%
Processing baselines	00.01 (16), 02.11 (22), 05.00 (6)	05.11 (49)
Distinct dates	19	25

Findings:

1. **Duplicates.** The same tile and date appears up to three times (reprocessed versions). Deduplication is now by tile and calendar date, keeping the highest baseline.
2. **Month imbalance.** T1 has no January 2019 scenes and only two February dates, so winter-crop greenness could differ between epochs for seasonal reasons. Month-balanced composites (shared months only) are now the default.
3. **Two UTM zones.** The area straddles 78°E. Data from zone 43 are reprojected to EPSG:32644 (bilinear for reflectance, nearest for SCL).
4. **Tile-level cloud cutoff dropped.** It describes a 110 km tile, not the 30 km study square. SCL pixel masking is used instead.

5.3 Reflectance harmonisation check (VERIFIED, partial)

Same tile (43RGP) and date (2018-12-01), baseline 00.01 vs. 05.00, 100×100 pixels at 10 m:

Band	00.01 mean	05.00 mean	Difference
B04 (red)	0.0771	0.0770	-0.0001
B08 (NIR)	0.2182	0.2194	+0.0012
B11 (SWIR)	0.1790	0.1798	+0.0008

Agreement is within about 0.001, so no spurious 0.1 offset was introduced for this case. **Limits:** one tile, one date, a 1 km window, and only items flagged `boa_offset_applied = True`. Items with the flag False or absent are **untested**. A script covering all scenes has been written and is **NOT YET RUN**.

5.4 Test suite on Colab (VERIFIED)

101 tests, 100 passed, 1 failed, 0 skipped. The raster tests that are skipped on Windows ran and passed on Linux. The one failure is `test_pre_and_post_2022_values_are_harmonized`: a post-2022 scene with no flag now gets 0.6 instead of 0.5 reflectance. This follows from the rule "trust the item's declared scale and offset". It does not affect the Earth Search run (all items declare an offset), but it must be resolved by a documented rule, not by editing the expected value.

6. Issues found and what was done

#	Issue	Action
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1	Pipeline never ran on real imagery; early reports said "complete"	Reports corrected; "NOT YET RUN" convention adopted
2	RF trained on rule-based pseudo-labels (circular)	Marked bootstrap only; manual labels and spatial split planned
3	Class scheme silently reduced to 4 classes	Restored to 5
4	AOI was 27×46 km, included Mussoorie, stored as a point	Replaced by the approved 30 km square polygon
5	Planned GIS library, PostGIS and manifest were skipped	Added in Phase 3
6	Sentinel-2 offset change after January 2022	Per-scene handling, partial real check done
7	Hansen tile named wrongly (30N_070E)	Corrected to 40N_070E
8	Mock benchmark's 1.00 scores could be mistaken for accuracy	Reports reworded; live benchmark script added
9	Rasterio blocked locally; remote reads failed	Moved to Colab
10	Notebook errors: Python 3.11 guard, nested repo folder, hidden subprocess errors, missing pystac	Notebook rewritten with captured output and root-finding
11	Duplicate scenes, uneven months, tile-level cloud filter	Dedup, month-balancing, SCL masking added

7. Verified vs. not yet run

Item	Status
Synthetic unit tests (GIS, indices, classifiers, validators, agent)	VERIFIED on Colab (100 of 101 pass)
STAC access, scene inventory, GDAL reads from public COGs	VERIFIED
Reflectance agreement across baselines (one window)	VERIFIED, partial
T1 and T2 composites, valid-pixel %, previews	NOT YET RUN
T1 vs. T2 brightness consistency, seam check	NOT YET RUN
Land-cover maps, transition matrix, change areas	NOT YET RUN
Reference labels, accuracy, kappa	NOT YET RUN (no labels exist)
Hotspots, Moran's I, Spearman result	NOT YET RUN
GHSL / Hansen / WorldCover comparison	NOT YET RUN (tiles not downloaded)
PostGIS queries against a live database	NOT YET RUN
Live LLM planner benchmark	NOT YET MEASURED

RAG on real papers

NOT YET RUN (no PDFs ingested)

Streamlit app launched and screenshotted

NOT YET VERIFIED

8. Known limitations

- 10 m resolution cannot resolve individual buildings or narrow roads. Results are settlement-scale.
- Built-up and bare soil are spectrally similar, which is a classic source of error.
- Hill shadows, seasonal vegetation and residual atmospheric effects can create false NDVI change.
- Post-classification change compounds the errors of two maps.
- Validation will cover one region and a modest number of labels.
- Hotspot p-values use an analytical normal approximation, not a field-validated test.
- Citation retrieval shows what was retrieved, not that a claim follows from it.

9. Remaining plan and who does what

#	Task	Who	Notes
1	Rerun offset logic across all scenes (including flag False/absent)	Agent writes, a teammate runs on Colab	Resolve the failing test with a documented rule
2	Smoke test, then T1 and T2 composites	Teammate on Colab	Stop at the first failure and share the traceback
3	Review previews, valid-pixel %, seams, T1/T2 brightness	Whole team	Look at images before trusting numbers
4	Label about 150 points	Two people, independently	Needed for kappa. Write a short labelling guide first
5	Run Phases 4–5 on real composites	Teammate on Colab	First real change map and hotspots
6	Download GHSL, Hansen 40N_070E, WorldCover tiles	Teammate	For consistency checks
7	Collect and verify 15–25 open-access papers	Whole team	Verify every citation by hand
8	Run the live LLM benchmark	Teammate with an API key	Have a teammate write 10 extra test queries
9	PostGIS check, Streamlit launch, screenshots	Teammate	Document if Docker is unavailable
10	Final evaluation, report and paper	Whole team	After steps 1–9

10. How to run the hosted workflow

1. Push the latest code to GitHub (check `git status` for `.venv`, `.env`, `data/`).
2. In Colab: Runtime > Restart session. Run the clone cell (it finds `pyproject.toml` one folder down), then the install cell.
3. Run the full pytest cell. Expect no failures once the offset rule is settled.
4. Run `check_offset.py`, then the smoke test (5×5 km, 3 scenes, one month), then T1 and T2 in separate cells.
5. Copy outputs to Google Drive. Paste versions, scene counts, valid-pixel % and any traceback for review.

Rule of thumb: anything raster-related is only real if it ran on Colab or Kaggle. The agent on the Windows machine cannot execute raster code.

11. Key terms

- **SCL**: Sentinel-2 Scene Classification Layer (clouds, shadows, snow, and so on).
- **COG**: Cloud-Optimized GeoTIFF.
- **NDVI / NDBI / MNDWI / BSI**: vegetation, built-up, water and bare-soil spectral indices.
- **MMU**: minimum mapping unit (smallest patch kept; 0.5 ha here).
- **Moran's I / Gi***: spatial autocorrelation and hotspot statistics.
- **Cohen's kappa**: agreement between two annotators beyond chance.
- **RAG**: retrieval-augmented generation (answers grounded in retrieved documents).